HVIA

OLIST ANALYSIS

Turning data into your strongest competitive advantage. Using Python, SQL, and Power BI, we build end-to-end pipelines that uncover critical insights and drive high-impact solutions for HVIA. Let's partner to scale your business today.

HAGER_ABDELNASSER

WELCOME

Hi Hager Abdelnasser , I'm an analyst at HVIA Data & AI Solutions. While studying Olist's public order data, one pattern stood out: only ~8% of deliveries were late, yet those orders averaged 2.6 stars versus 4.3 for on-time ones, and lateness clustered in peak months and in the North and Northeast.

We help e-commerce teams turn this kind of signal into delivery-risk prediction and more accurate delivery promises. I'd be glad to share a short analysis and hear how your team handles this today. Would you be open to a 20-minute call in the next couple of weeks?
Best, [Your name]



AGENDA&APPROACH



1. Technical Pipeline:
ETL (Python) ==>
Constraints & Star Schema Integrity (SQL) ==>
Insights & Dashboards (Power BI)

2. Business Dashboard & Analysis:
Walking through our interactive dashboards to showcase deep-dive insights, pinpointing critical business challenges, and highlighting core operational strengths

3. HVIA Tailored Solutions: Presenting actionable, data-driven solutions and strategic recommendations designed for HVIA to optimize performance and maximize growth

ETL

1. Extract: Ingested 9 raw Olist e-commerce datasets simultaneously into a dynamic Python dictionary using pd.read_csv(), laying the foundation for pipeline automation.

2. Transform: Cleaned data by dropping duplicates, handling missing values, standardizing states to uppercase, parsing timestamps into datetimes, translating product categories, and structuring a Star Schema (dim_customers, dim_products, dim_sellers, dim_date, and fact_order_items).

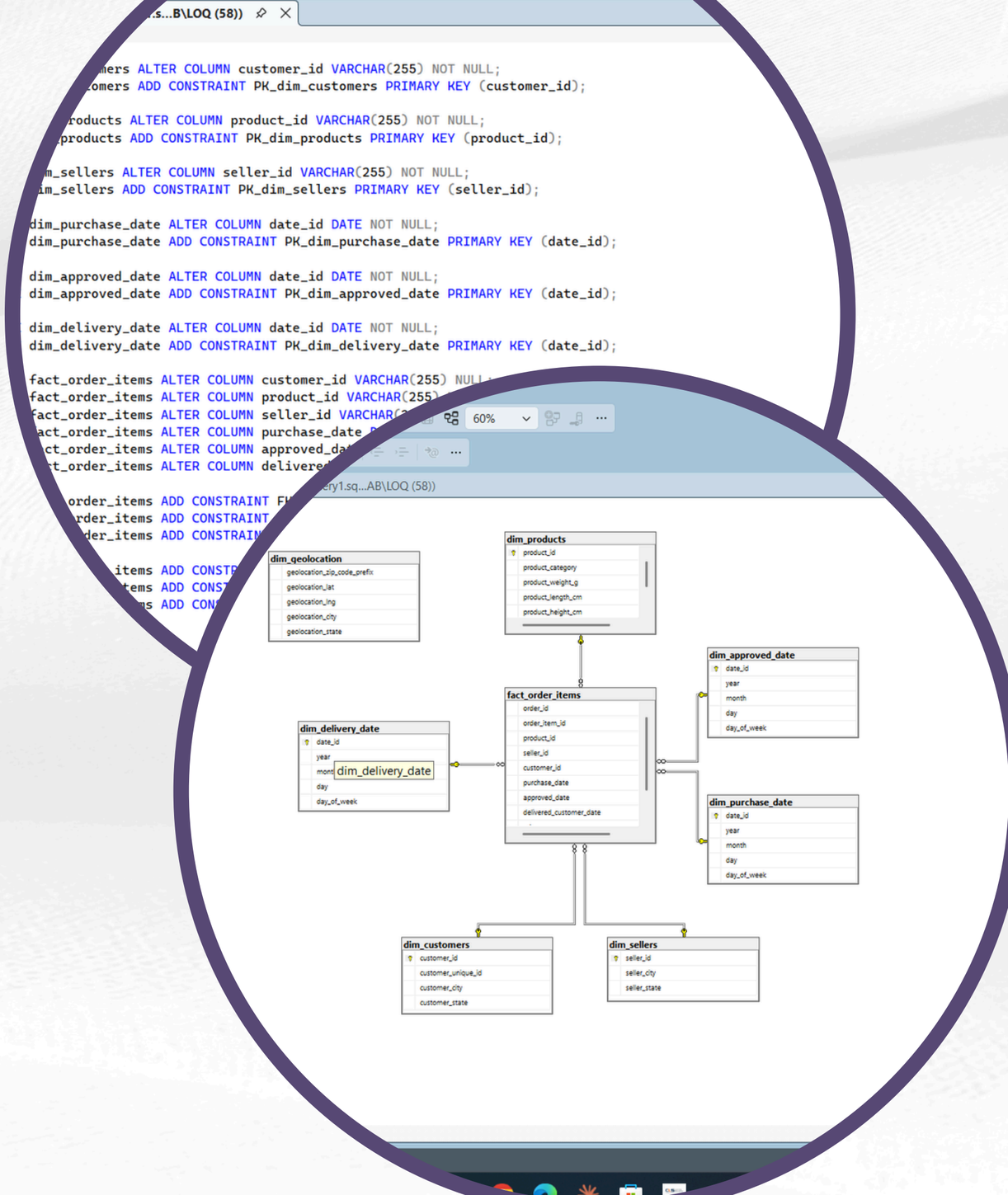
3. Load: Established a secure connection with SQL Server via SQLAlchemy and ODBC Driver 17, successfully persisting all cleaned dimension and fact tables into the Olist_DW database (if_exists='replace').



CONSTRAINTS & STAR SCHEMA INTEGRITY (SQL)

1. Data Integrity & Constraints: Enforced NOT NULL constraints on primary keys and added Foreign Key (FK) relationships linking fact_order_items to dimension tables (dim_customers, dim_products, dim_sellers, and dim_date) to ensure referential integrity, prevent orphan records, and optimize analytical queries

2. Star Schema Architecture: Successfully structured the final Olist_DW database into a classic Star Schema in SQL Server Management Studio (SSMS), placing the transaction table (fact_order_items) at the center surrounded by clean, related dimension tables for seamless reporting.



POWER BI MODELING & VISUALIZATION WORKFLOW

Semantic Model & Relationship Setup: Connected the cleaned Olist Star Schema tables (fact_order_items, dim_customers, dim_products, dim_sellers, and date dimensions) within Power BI, establishing robust one-to-many relationships to ensure accurate metric evaluation and cross-filtering.

DAX Measures & Advanced Metrics: Developed custom DAX measures (such as delivery time calculations using DATEDIFF and AVERAGEX, along with aggregations for order costs and payment behaviors) to transform raw relational data into actionable executive insights.

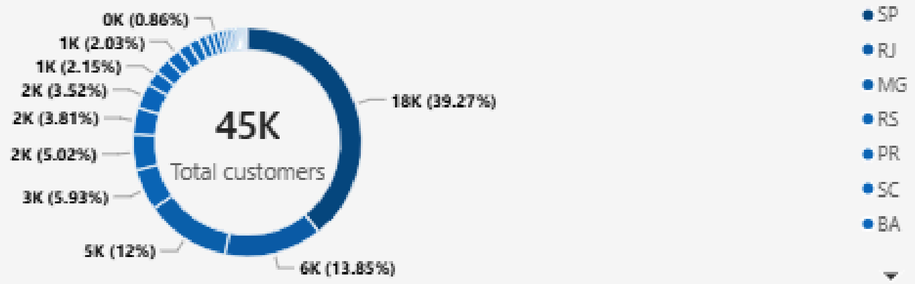


Customer Analytics Dashboard

Total cus...	Total Sates
45K	4K

year: 2016 – 2017

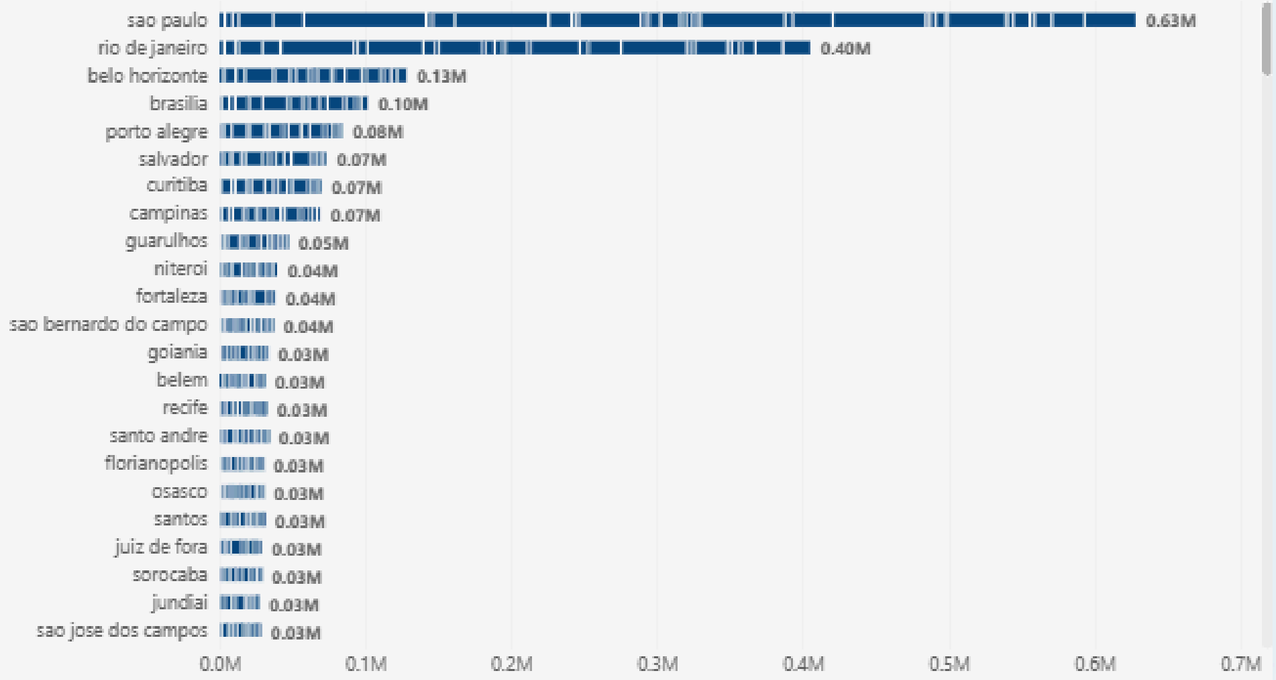
Total customers
by customer_state



Total customers
by Year, Quarter, Month and Day

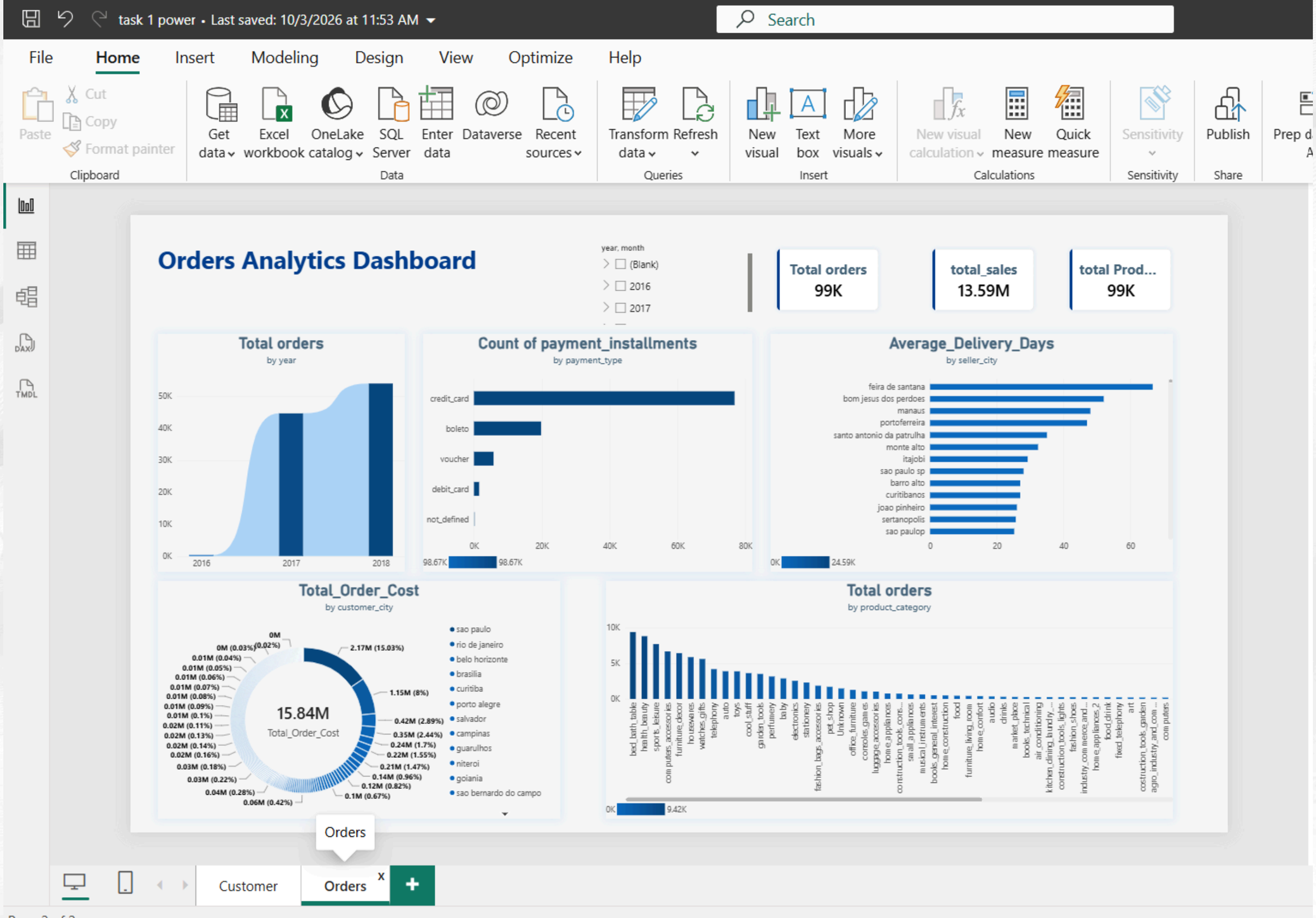


Total_Order_Cost
by customer_city and product_category



Count of geolocation_lng, Average of geolocation_lng and Total customers
by geolocation_zip_code_prefix and geolocation_city




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Search
File
Home
Insert
Modeling
Design
View
Optimize
Help
Paste
Cut
Copy
Format painter
Clipboard
Get data
Excel workbook
OneLake catalog
SQL Server
Enter data
Dataverse
Recent sources
Data
Transform data
Refresh
Queries
New visual
Text box
More visuals
Insert
New visual calculation
New measure
Quick measure
Calculations
Sensitivity
Sensitivity
Publish
Share
Orders Analytics Dashboard
year, month
(Blank)
2016
2017
Total orders
99K
total_sales
13.59M
total Prod...
99K
Total orders
by year
50K
40K
30K
20K
10K
0K
2016
2017
2018
Count of payment_installments
by payment_type
credit_card
boleto
voucher
debit_card
not_defined
0K
20K
40K
60K
80K
98.67K
98.67K
Average_Delivery_Days
by seller_city
feira de santana
bom jesus dos perdoes
manaus
portoferreira
santo antonio da patrulha
monte alto
itajobi
sao paulo sp
barro alto
curitibanos
joao pinheiro
sertanopolis
sao paulop
0
20
40
60
0K
24.59K
Total_Order_Cost
by customer_city
15.84M
Total_Order_Cost
2.17M (15.03%)
1.15M (8%)
0.42M (2.89%)
0.35M (2.44%)
0.24M (1.7%)
0.22M (1.55%)
0.21M (1.47%)
0.14M (0.96%)
0.12M (0.82%)
0.1M (0.67%)
0.06M (0.42%)
0.04M (0.28%)
0.03M (0.22%)
0.03M (0.18%)
0.02M (0.16%)
0.02M (0.14%)
0.02M (0.13%)
0.02M (0.11%)
0.01M (0.1%)
0.01M (0.09%)
0.01M (0.08%)
0.01M (0.07%)
0.01M (0.06%)
0.01M (0.05%)
0.01M (0.04%)
0M (0.03%)
0M (0.02%)
sao paulo
rio de janeiro
belo horizonte
brasilia
curitiba
porto alegre
salvador
campinas
guarulhos
niteroi
goiania
sao bernardo do campo
Total orders
by product_category
10K
5K
0K
bed_bath_table
health_beauty
sports_leisure
computers_accessories
furniture_decor
housewares
watches_gifts
telephony
auto
toys
cool_stuff
garden_tools
perfumery
baby
electronics
stationery
fashion_bags_accessories
pet_shop
Unknown
office_furniture
consoles_games
luggage_accessories
home_appliances
construction_tools_cons...
small_appliances
musical_instruments
books_general_interest
home_construction
food
furniture_living_room
home_comfort
audio
drinks
market_place
books_technical
air_conditioning
kitchen_dining_laundry_...
construction_tools_lights
fashion_shoes
industry_commerce_and...
home_appliances_2
food_drink
fixed_telephony
art
construction_tools_garden
agro_industry_and_com...
computers
0K
9.42K
Orders
Customer
Orders
DAX
TMDL

FINDING

Finding 1 :
Rapid growth in 2017, then a plateau
Monthly orders rose from about 800 (Jan 2017) to 7,544 in Nov 2017 (Black Friday), then settled at roughly 6,200–7,300 per month through Aug 2018. Monthly sales peaked near R$1.0M. The early growth engine had stopped compounding by 2018.

Finding 2 :
Late deliveries are the biggest satisfaction problem

- 8.1% of delivered orders arrived after the estimated date.
- Average review score: 4.29 for on-time orders vs 2.57 for late orders.
- 54% of late orders get 1–2 stars, versus 9% of on-time orders.
- Score falls steadily with delivery time: 4.41 (≤7 days), 4.10 (15–21 days), 3.49 (22–30 days), 2.17 (over 30 days).
- Late rates spiked at peak or stress periods: 14.3% (Nov 2017), 16.0% (Feb 2018), 21.4% (Mar 2018).

If late orders were rated like on-time ones, the overall average would rise by about 0.14 points (4.09 to about 4.23). Treat that as an upper bound, because some low scores come from product problems rather than lateness.

FINDING

Finding 3 :
Geography: a Southeast-centred marketplace with long tails

- 42% of customers are in São Paulo (SP), then Rio (13%) and Minas Gerais (12%). 64% of seller sales come from SP sellers.
- 64% of order items cross state lines, so logistics performance matters across the whole system.
- Delivery is much slower and less reliable outside the core: SP averages 8.3 days, RJ 14.9 days (13% late), and northern or northeastern states such as AL (24 days, 24% late) and MA (21 days, 20% late) are the weakest.

Finding 4 :
Sales are concentrated in a few sellers
The top 10% of sellers generate 67% of sales and the top 1% generate 26%. The median seller sold only about R$824 over the whole period. That implies a large tail of sellers who barely activate, a retention and onboarding problem and a concentration risk.

FINDING

Finding 5 :
Customers rarely come back
Only 3.1% of unique customers placed more than one order. Orders are small (average R$137, median R$87, 1.14 items per order), and only 1.3% involve more than one seller. Acquisition drives growth, not loyalty.
Finding 6 – Freight is a heavy cost for bulky categories
Freight equals 16.6% of item value overall (median order about 22%). It reaches about 24% for furniture/decor and 23% for housewares, versus 8% for watches and gifts. Top categories by sales are health/beauty, watches/gifts, bed/bath/table, sports and computer accessories.
4. Solution Propos

WHAT HVIA COULD OFFER

4. Solution Proposal: What HVIA Could Offer

Lead idea: Delivery Risk and Promise Engine (Logistics Intelligence).

Problem worth solving. 8% of orders are late and those orders account for most of the poor reviews. Late rates spike in specific months and regions, which suggests the delivery promise is not calibrated to actual conditions.

What HVIA would build. A model that predicts, at checkout or at dispatch, the probability that an order will miss its promised date, using seller, product weight and size, origin-destination route and season. It feeds (1) more accurate delivery promises shown to the buyer, (2) early alerts for at-risk orders so support can intervene, and (3) a seller and carrier scorecard.

Evidence. Findings 2 and 3: a clear satisfaction cost per late order, with predictable regional and seasonal patterns.

Success metrics. Late-delivery rate, share of 1–2 star reviews, average review score, support contacts per order.

Follow-on opportunities

• Seller health and activation analytics (Analytics and Decision Systems). Score sellers on activity, delivery quality and reviews. Target the long tail for activation and the top 10% for retention. This ties into credit-risk decisions for Olist's lending products.

• Customer intelligence and retention (Customer Intelligence). With only 3% repeaters, a segmentation and next-best-offer model for repeat purchase could be valuable, though marketplace data limits what Olist can see of the buyer.

• Freight and category pricing (Forecasting and Predictive Solutions). Show sellers when freight makes a product uncompetitive and forecast demand peaks such as Black Friday.

Caveats to validate with Olist: the data is from 2016–2018, payments data was not available here, and the relationships above are correlations rather than proven causes.



HVIA

Ready for Smarter Data, Better Decisions, Bigger Results?

Start Your Journey with HVIA.

THANK YOU

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